

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5123C

5 Credits: 4

Program Elective

Source-grounded

Process Control And

□ To impart knowledge on measurement of various parameters and instrumentation for

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5123C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5123C.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5123C
Course title	Process Control And
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge on measurement of various parameters and instrumentation for
- processes in wood based industries.
- To provide knowledge of process control.

Course outcomes

CO	Official outcome	Study level
CO1	static characteristics of instruments in wood based 15 Understanding industries Describe the working principle of measuring	See official syllabus
CO2	14 Understanding instruments in process plants Outline the measuring instruments in paper	See official syllabus
CO3	15 Understanding industry Explain the modes of control action and the	See official syllabus
CO4	14 Understanding computer control system	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	instruments in wood based industries.	4	As specified
Module 2	plants	4	As specified
Module 3	Outline the measuring instruments in paper industry	3	As specified
Module 4	Explain the modes of control action and the computer control system	4	As specified

MODULE 1

instruments in wood based industries.

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: instruments in wood based industries.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding industries

1 Understanding industries is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding industries using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding industries should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding industries means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding industries

പ്രദേശം/സ്ഥലം definition/meaning പ്രദേശം/സ്ഥലം. പ്രദേശം/സ്ഥലം പ്രദേശം/സ്ഥലം, steps, application പ്രദേശം/സ്ഥലം പ്രദേശം/സ്ഥലം. Technical terms English- പ്രദേശം/സ്ഥലം പ്രദേശം/സ്ഥലം.

▼ Construct the instrumentation block diagram

Construct the instrumentation block diagram is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Construct the instrumentation block diagram using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, construct the instrumentation block diagram should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Construct the instrumentation block diagram means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Construct the instrumentation block diagram definition/meaning. Steps, application. Technical terms English-Construct the instrumentation block diagram.

▼ Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic

Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, suggest the functions of measuring instruments 6 understanding analyze different static and dynamic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic definition/meaning, steps, application Technical terms English-

▼ 7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process

7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Understanding characteristics of instruments
Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 understanding characteristics of instruments importance of instrumentation in process industries; block diagram; functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. transducers, their classification and selection criteria; on and offline measurements. static characteristics- accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone dynamic characteristics, speed of response, fidelity, lag, dynamic error. describe the working principle of measuring instruments in process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of

പ്രക്രിയ നിയന്ത്രണത്തിന്റെ. പദപദപദം പദപദപദം, steps, application പദപദപദം
പദപദപദപദം പദപദപദം. Technical terms English-പദ പദപദ പദപദപദപദപദപദ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**plants**

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: plants
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding measuring instruments Discuss the working principle of Pressure

3 Understanding measuring instruments Discuss the working principle of Pressure is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding measuring instruments Discuss the working principle of Pressure using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding measuring instruments discuss the working principle of pressure should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding measuring instruments Discuss the working principle of Pressure means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding measuring instruments Discuss the working principle of Pressure definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 3 Understanding measuring instruments Outline the working principle of flow and level

3 Understanding measuring instruments Outline the working principle of flow and level is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding measuring instruments Outline the working principle of flow and level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding measuring instruments outline the working principle of flow and level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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▼ 3 Understanding measuring instruments Illustrate the working principle of PH, viscosity

3 Understanding measuring instruments Illustrate the working principle of PH, viscosity is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding measuring instruments Illustrate the working principle of PH, viscosity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding measuring instruments illustrate the working principle of ph, viscosity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding measuring instruments Illustrate the working principle of PH, viscosity means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding measuring instruments
Illustrate the working principle of PH, viscosity definition/meaning. Explain the steps, application of PH. Technical terms English-Malayalam.

▼ 5 Understanding and humidity measuring instruments

Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer.

5 Understanding and humidity measuring instruments

Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture-

infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer. is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding and humidity measuring instruments
Temperature measuring instruments - Principle and working of - Thermometers, filled system and vapor pressure thermometers - Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture-infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding and humidity measuring instruments temperature measuring instruments - principle and working of -thermometers, filled system and vapor pressure thermometers -thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. pressure measuring instruments - principle and working of - bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. flow measuring devices: coriolis mass flow meter, ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. level measuring devices: float level indicator, magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. ph measurement, ph scale, reference and measuring electrode - conductivity measurement, electrical conductivity meter - density measurement, online density meter, hydrometer. humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type viscosity - absolute and kinematic viscosity - viscosity measurement - say bolt and redwood viscometer - falling- ball viscometer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of - Thermometers, filled system and vapor pressure thermometers - Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity

Aspect	What to prepare
	measurement, electrical conductivity meter – Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers - Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter – Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic

viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer. വിശിഷ്ടതാപകരണത്തിന്റെ നിർവ്വചനം/അർത്ഥം വിശദീകരിക്കുക. വിശിഷ്ടതാപകരണത്തിന്റെ ഘട്ടങ്ങൾ, പ്രയോഗങ്ങൾ വിശദീകരിക്കുക. Technical terms English- Malayalam വിശദീകരിക്കുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Outline the measuring instruments in paper industry

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the measuring instruments in paper industry
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 5 Understanding paper industries

5 Understanding paper industries is an official topic in Module 3 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding paper industries using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding paper industries should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding paper industries means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding paper industries
 പേപ്പർ ഇൻഡസ്ട്രിയിൽ നിന്ന് definition/meaning നൽകുക. പേപ്പർ ഇൻഡസ്ട്രിയിൽ
 ഉപയോഗിക്കുന്ന steps, application നൽകുക. Technical terms
 English-ൽ നൽകുക.

▼ Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement

Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement is an official topic in Module 3 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the properties of paper 5 understanding illustrate the working principle for measurement should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement
 പരീക്ഷയിൽ നിങ്ങൾക്ക് definition/meaning നൽകേണ്ടതാണ്. അതിനായി പരീക്ഷയിൽ
 പരീക്ഷിക്കുക, steps, application നൽകുക പരീക്ഷിക്കുക. Technical terms
 English-ൽ പരീക്ഷിക്കുക പരീക്ഷിക്കേണ്ടതാണ്.

▼ 5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper

5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper is an official topic in Module 3 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water

absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding of properties of paper definitions and methods of determination of : physical properties - grammage, caliper, bulk, smoothness and porosity; mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: optical properties: brightness, opacity, gloss and color: resistance properties: water absorptiveness, cobb test, oil absorbency, air/gas permeability. electrical properties: dielectric constant, dielectric strength, dielectric loss and properties of electrical insulating paper should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding of properties of paper
Definitions and methods of determination of : Physical properties -
Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile
strength, tear strength, burst strength, folding endurance, and bending
stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance
Properties: water absorptiveness, Cobb test, oil absorbency, air/gas
permeability. Electrical properties: Dielectric constant, Dielectric strength,
Dielectric loss and Properties of electrical insulating paper
definition/meaning . steps,
application . Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain the modes of control action and the computer control system

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the modes of control action and the computer control system
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding control.

3 Understanding control. is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding control. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 3 Understanding control. പദപരിഭാഷണം പദപരിഭാഷണം definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-ൽ പദപരിഭാഷണം പദപരിഭാഷണം.

▼ Comprehend computerized control system.

Comprehend computerized control system. is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend computerized control system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend computerized control system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comprehend computerized control system. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓം Comprehend computerized control system. ഓംഓംഓംഓംഓംഓം ഓംഓം definition/meaning ഓംഓംഓംഓംഓംഓം. ഓംഓംഓം ഓംഓംഓം, steps, application ഓംഓംഓം ഓംഓംഓംഓംഓം ഓംഓം. Technical terms English-ഓം ഓംഓം ഓംഓംഓംഓംഓം.

▼ Summarize final control elements.

Summarize final control elements. is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize final control elements. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize final control elements. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize final control elements. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ **Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1 <https://www.nptel.ac.in/courses/103103037/> [https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/](https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/)**

Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers - proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants -

Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1 <https://www.nptel.ac.in/courses/103103037/> [https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/](https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/) is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed-back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata McGraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1 <https://www.nptel.ac.in/courses/103103037/> [https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/](https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline instrumented safety system. 4 understanding the control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. outline computerized control in process plants - programmable logic controllers - supervisory control and data acquisition (scada) - rtu (remote terminal unit) - distributed control system (dcs). final control elements - mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, i to p converter. illustrate valve safe positions, fail open and fail close. outline fail steady position and variable frequency drive [vfd] instrumented safety system- emergency shutdown system - safety interlocks. text / reference: t/r book title/author t1 eckman, d.p., industrial instrumentation, wiley eastern ltd., new york, 1990. r1 patranabis, principles of industrial instrumentation, tata mcgraw hill, 2008. ghosh a.k., "introduction to instrumentation and control", 4th ed., prentice r2 hall of india. casey j.p., "pulp and paper chemistry and chemical technology", 3rd ed., r3 vol 3, john wiley & sons. r4 robert j' measurement and control in paper making' mc grill mark r. e., "handbook of physical and mechanical testing of paper and r5 paperboard", vol. 1, marcel dekker. sankaranarayanan, p.e., "pulp and paper industries technology and r6 instrumentation", kotharis desk book series, 1995. g. stephanopoulous, 'chemical process control - theory and practice', r7 prentice hall of india ltd. r8 lecture notes ipir, banglore online resources: sl.no website link 1 <https://www.nptel.ac.in/courses/103103037/> <https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed

Aspect	What to prepare
	control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous. 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1 https://www.nptel.ac.in/courses/103103037/ https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog

controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: SI.No Website Link 1 <https://www.nptel.ac.in/courses/103103037/> <https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/> നിർവ്വചന/അർത്ഥം നിർവ്വചനം. നിർവ്വചനം നിർവ്വചനം നിർവ്വചനം, steps, application നിർവ്വചനം നിർവ്വചനം നിർവ്വചനം. Technical terms English-ന നിർവ്വചനം നിർവ്വചനം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain 1 Understanding industries. (3 marks)

► Define or explain Construct the instrumentation block diagram. (3 marks)

► Define or explain Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic. (3 marks)

► Define or explain 7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process. (3 marks)

Module 2

► Define or explain 3 Understanding measuring instruments Discuss the working principle of Pressure. (3 marks)

► **Define or explain 3 Understanding measuring instruments Outline the working principle of flow and level. (3 marks)**

► **Define or explain 3 Understanding measuring instruments Illustrate the working principle of PH, viscosity. (3 marks)**

► **Define or explain 5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer.. (3 marks)**

Module 3

► **Define or explain 5 Understanding paper industries. (3 marks)**

► **Define or explain Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement. (3 marks)**

► **Define or explain 5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper. (3 marks)**

Module 4

► **Define or explain 3 Understanding control.. (3 marks)**

► **Define or explain Comprehend computerized control system.. (3 marks)**

► **Define or explain Summarize final control elements.. (3 marks)**

► **Define or explain Outline instrumented safety system. 4**
Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh

**A.K., “Introduction to Instrumentation and Control”, 4th Ed., Prentice R2 Hall of India. Casey J.P., “Pulp and Paper Chemistry and Chemical Technology”, 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J’ Measurement and control in paper making’ Mc Grill Mark R. E., “Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard”, Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., “Pulp and Paper Industries Technology and R6 Instrumentation”, Kotharis Desk book series, 1995. G. Stephanopoulous, ‘Chemical Process Control - Theory and Practice’, R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1
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[https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf- 2 notes-ics-pdf-notes/](https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/). (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding industries**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding measuring instruments Discuss the working principle of Pressure**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Understanding paper industries**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **3 Understanding control..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

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QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.nptel.ac.in/courses/103103037/>
- <https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-notes-ics-pdf-notes/>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5123C. Verify institutional updates against the current official curriculum.